

Baseline Summary — Vine Copula

Statistical dependence-model baseline for synthetic EV charging session generation (arrival hour, plug-in duration, energy).

1. The idea: separate the marginals from the dependence

A copula is a tool for building a joint distribution out of (i) each variable's own marginal distribution and (ii) a separate object — the copula — that encodes only how the variables move together. This split is **Sklar's theorem**: any joint CDF factorises as

$$F(a, d, e) = C(F_A(a), F_D(d), F_E(e))$$

where F_A, F_D, F_E are the marginals of arrival, duration and energy, and C is a copula on the unit cube $[0,1]^3$. The point: you can model the shape of each variable *and* the dependence between them independently, then recombine. This is attractive for EV data, where the marginals are strongly non-Gaussian and the dependence (e.g. late arrival → shorter connection; longer duration → more energy) matters.

2. Why 'vine': pair-copula construction

A single multivariate copula (e.g. Gaussian or Student-t) forces one global dependence shape on all variable pairs. A **vine copula** instead decomposes the multivariate dependence into a cascade of *bivariate* copulas arranged in a nested tree structure (a “regular vine”, R-vine; special cases C-vine / D-vine). For three variables the density factorises as

$$c(u_A, u_D, u_E) = c_{AD} \cdot c_{AE} \cdot c_{DE|A}$$

i.e. two unconditional pair-copulas (arrival–duration, arrival–energy) plus one *conditional* pair-copula (duration–energy given arrival). Crucially, **each pair can use a different copula family** (Gaussian, Student-t, Clayton, Gumbel, Frank, ...), so the model can represent heterogeneous and asymmetric dependence — including *tail dependence* (joint extremes), which a single Gaussian copula cannot.

3. How it is used as a generator (fit → sample)

- **Fit marginals.** Estimate F_A, F_D, F_E from the real sessions (empirical CDF or parametric), and map each observation to a uniform pseudo-observation $u = F(x)$.
- **Select & fit the vine.** Choose the tree structure and, for every edge, the best copula family + parameters by maximum likelihood with AIC/BIC selection (and Kendall's-tau inversion for the Archimedean families).
- **Sample.** Draw uniforms from the fitted vine (sequential conditional sampling down the trees), then invert the marginals $x = F^{-1}(u)$ to get synthetic (arrival, duration, energy) tuples.

4. Strengths and limitations

Strengths	Limitations
Data-fitted; fits our “fit → generate → compare marginals” protocol exactly.	Static i.i.d. sessions: no per-user identity and no sequential / idle-gap structure (like GMMNet and EV-SDG).
Best-in-class at <i>dependence</i> : per-pair families capture nonlinear, asymmetric and tail dependence a Gaussian copula misses.	With only 3 variables the vine's high-dimensional advantage is modest; most of the gain is per-pair family flexibility.
Interpretable and robust in low-data regimes (strong on small cohorts in the 2026 EV study).	Transductive: reproduces the training population, does not individualise or extrapolate a timeline.

5. Role in our study & EV precedent

The vine copula is the **statistical dependence-model** baseline, complementing the parametric EV-SDG and the neural GMMNet. It is well precedented for EV charging events: Einolander & Lahdelma (*Energy*, 2022) applied multivariate copulas to arrival/duration/energy and found the Student-t copula most reliable; Výboh & Grmanová (2026) introduced **Vine copulas + CODINE** to the EV domain on exactly these three variables and on the same Norway (Trondheim/Sørensen) dataset we use, reporting that vine copulas best preserve tail and correlation structure and are the most stable in low-data settings. Implementation is straightforward with `pyvinecopu.lib` (Python).

Expected behaviour in our comparison: like GMMNet, it should match the real marginals well and be *strong on the dependence metrics* (Kendall-tau, tail dependence) — possibly stronger than our method there. Like the other baselines, it produces independent sessions, so it cannot model the idle gap or a per-user timeline — keeping the sequential structure as our method's distinguishing contribution.